

Renewable Intermittency and Grid Reliability

A Directed Technical Change DSGE Model of Vietnam

Midterm Progress Report

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Presentation Outline

- 1 Motivation & Research Questions
- 2 Model Environment
- 3 Calibration
- 4 Model Mechanics & Log-Linearization
- 5 Preliminary Results
- 6 Work in Progress & Timeline

Vietnam's Energy Trilemma

Vietnam's growth model confronts three simultaneously binding constraints:

- **Energy security:** rapidly rising industrial demand
- **Sustainability:** Paris Agreement + Net Zero pledges
- **Affordability:** administratively regulated retail tariffs (EVN)

PDP8 at a Glance

	2024 actual	2030 target
Solar + Wind	22.8 GW	73.5 GW
Battery BESS	0.25 GW	14.0 GW
Grid invest.	\$3.2 B/yr	\$3.6 B/yr

The Core Structural Friction

Renewable generation is *non-dispatchable*: supply clears in real time with demand. As VRE penetration rises without adequate storage, the **reliability penalty**—endogenous reduction in capital utilization from grid instability—acts like a persistent negative TFP shock.

Empirical Trigger

Solar capacity doubled in 18 months (2020–21) under FiT incentives, while battery storage remained near zero. Vietnam already experiences curtailment $>10\%$ in southern provinces.

Literature Gap: What Existing Models Miss

What the literature covers:

- *E-DSGE models* (Heutel 2012; Airaud et al. 2025): carbon taxes, “greenflation”
- *Directed Technical Change* (Acemoglu et al. 2012): R&D-race between sectors
- *Techno-economic* (Zakeri & Syri 2015; Schreiner 2020): storage costs, grid investment
- *Open economy* (Schmitt-Grohé 2003): debt-elastic interest rates

Critical blind spots:

- No model *endogenizes grid reliability* as a capital utilization constraint
- DTC literature ignores *deployment-driven learning* in storage
- Public–private investment *asymmetry* (“agility gap”) is not formalized
- Open-economy E-DSGE ignores *battery import price* as a terms-of-trade shock

This capstone bridges all four gaps jointly—in a single DSGE calibrated to Vietnam.

Research Questions & Contributions

Four Research Questions

- ① How does VRE intermittency propagate through capital utilization into macroeconomic output losses?
- ② What is the structural timing mismatch between private battery and public grid investment—the **Agility Gap**?
- ③ How does endogenous learning-by-doing in storage depend on regulatory price signal transmission (χ)?
- ④ What are the welfare costs of intermittency and how do policy instruments rank in closing the gap?

Contribution 1

Endogenous reliability function:
 $u_t = f(\text{flexibility/VRE vol})$

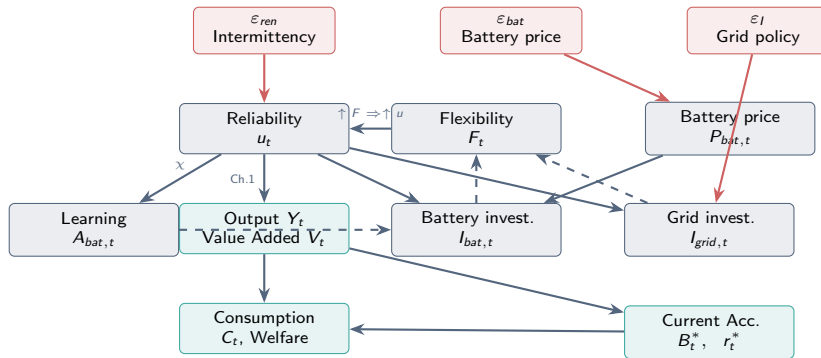
Contribution 2

Reliability-gap learning: ΔA_{bat}
 responds to scarcity signal χ

Contribution 3

CES agility gap: private batteries (agile) vs. public grid (sluggish)

Model Architecture: Small Open Economy DSGE



Solid = contemporaneous effect; *dashed* = dynamic feedback through capital accumulation.

Households

A representative household maximizes discounted lifetime utility:

$$\max_{\{C_t, L_t, K_{p,t+1}, B_t^*\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \frac{L_t^{1+\sigma_L}}{1+\sigma_L} \right] \quad (1)$$

subject to the **consolidated budget constraint**:

$$C_t + I_{p,t} + P_{bat,t} I_{bat,t} + B_t^* + T_t = Y_t + (1 + r_{t-1}^*) B_{t-1}^* \quad (2)$$

Capital Euler Equation

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{\alpha Y_{t+1} / K_{p,t+1} + 1 - \delta_p}{C_{t+1}} \right]$$

Labor Supply

$$(1 - \alpha) V_t = \varphi C_t L_t^{1+\sigma_L}$$

Key modeling choice: Capital utilization u_t is *not* a household decision variable. It reflects the fraction of installed capital that can be operated given grid reliability—an exogenous physical constraint, not a margin of adjustment.

Production Technology & Endogenous Reliability

Final output (CES of value-added and energy):

$$Y_t = \left[s_V V_t^{(\sigma_E-1)/\sigma_E} + (1 - s_V) \bar{E}^{(\sigma_E-1)/\sigma_E} \right]^{\sigma_E/(\sigma_E-1)}, \quad \sigma_E = 0.6$$

Value-added (Cobb-Douglas with endogenous utilization):

$$V_t = (u_t K_{p,t})^\alpha L_t^{1-\alpha}, \quad \alpha = 0.35$$

Reliability Constraint (Novel)

$$u_t = 1 - \exp\left(-\psi \frac{F_t}{\phi_{int} \cdot \text{Vol}_{ren,t}}\right)$$

When $F_t \uparrow$ (more flexibility) or $\text{Vol}_{ren,t} \downarrow$, utilization rises toward 1. The exponential form captures the *nonlinear* costs of reliability shortfalls.

CES Flexibility Bundle (Novel)

$$F_t = \left[\mu (A_{bat,t} K_{b,t})^{\frac{\rho_f-1}{\rho_f}} + (1-\mu) K_{g,t}^{\frac{\rho_f-1}{\rho_f}} \right]^{\frac{\rho_f}{\rho_f-1}}$$

Batteries and grid are *complements* ($\rho_f = 0.4 < 1$): neither substitutes for the other when the opposing asset is scarce.

Battery Investment Rule & Learning-by-Doing

Battery Investment (Private, Agile)

$$\frac{I_{bat,t}}{\bar{I}_{bat}} = \left(\frac{u^*}{u_t} \right)^{\phi_{grid}} \frac{1}{P_{bat,t}}$$

- Responds *contemporaneously* to reliability gap
- Dampened by world battery price $P_{bat,t}$
- Capital accumulates:
 $K_{b,t+1} = (1 - \delta_b)K_{b,t} + I_{bat,t}$

Endogenous Learning (Novel)

$$A_{bat,t} - A_{bat,t-1} = \eta_{bat} \cdot \chi \cdot \frac{u^* - u_t}{u^*}$$

- $\eta_{bat} = 0.10$: learning speed (18–22% Li-ion experience curve)
- $\chi \in [0, 1]$: **regulatory transmission parameter**
- $\chi = 1$: full price pass-through (liberalized market)
- $\chi = 0$: fixed tariffs suppress scarcity signal

The Agility Gap mechanism: When $u_t < u^*$, battery investment responds *immediately* (private) while grid investment requires multi-year construction lags (public). This differential speed is the source of the

Public Grid Investment & External Sector

Grid Investment Rule (Government)

$$\hat{l}_{grid,t} = -\phi_{grid} \hat{u}_t + \varepsilon_{l,t}, \quad \phi_{grid} = 1.5$$

- *Reactive policy*: invests more when reliability falls
- Lump-sum tax T_t finances grid; Ricardian equivalence holds
- $\varepsilon_{l,t}$: implementation delay shock (budget cycles, permitting)
- Capital: $K_{g,t+1} = (1 - \delta_g)K_{g,t} + I_{grid,t}$

Small Open Economy (Schmitt-Grohé 2003)

Current account / resource constraint:

$$B_t^* = (1 + r_{t-1}^*)B_{t-1}^* + Y_t - C_t - I_{p,t} - P_{bat,t}I_{bat,t} - I_{grid,t}$$

Debt-elastic interest premium:

$$r_t^* = \bar{r} + \phi_b \left(e^{\bar{B}^* - B_t^*} - 1 \right)$$

- $\phi_b = 0.001$ ensures stationarity of B^*
- Battery imports at world price $\Rightarrow$ terms-of-trade shock channel

Exogenous Shocks & Equilibrium Definition

Three Exogenous Driving Forces

$$\hat{E}_{ren,t} = \rho_{ren} \hat{E}_{ren,t-1} + \varepsilon_t^{ren}, \quad \varepsilon^{ren} \sim \mathcal{N}(0, \sigma_{ren}^2)$$

$$\hat{P}_{bat,t} = \rho_{bat} \hat{P}_{bat,t-1} + \varepsilon_t^{bat}, \quad \varepsilon^{bat} \sim \mathcal{N}(0, \sigma_{bat}^2)$$

$$\varepsilon_{I,t} = \rho_I \varepsilon_{I,t-1} + \nu_{I,t}, \quad \nu_I \sim \mathcal{N}(0, \sigma_I^2)$$

- All shocks mutually independent
- $\rho_{ren} = 0.85$ (monsoon seasonality)
- $\sigma_{ren} = 0.12$ (CV of monthly VN solar output)
- $\rho_{bat} = 0.90$, $\sigma_{bat} = 0.08$ (lithium price cycle)

Competitive Equilibrium: 16 Equations, 16 Unknowns

Block	Equations
Households	Capital Euler, Bond Euler, Labor supply
Production	CES output, Value-added
Energy/Grid	Reliability u_t , Flexibility F_t
Battery	I_{bat} rule, K_b accumulation, LBD A_{bat}
Grid	I_{grid} rule, K_g accumulation
Capital	K_p accumulation
External	Current account, Interest premium
Shocks	Battery price AR(1)

Blanchard-Kahn: 3 eigenvalues outside the unit circle, matching 3 forward-looking variables (C_t , Y_t , r_t^*). Unique saddle-path equilibrium confirmed.

Calibration Strategy: Three-Stage Approach

- 1 **Standard structural parameters** matched to long-run Vietnamese macroeconomic aggregates (GSO 2024, ADB, World Bank)
- 2 **Reliability & intermittency parameters** micro-founded on engineering data: NLDC facility-level generation, IEA Vietnam Energy Outlook 2023, EVN annual reliability reports
- 3 **Learning-by-doing parameters** anchored to empirical experience curve literature for Li-ion batteries (Kittner et al. 2017; Way et al. 2022)

Calibration Target: Flexibility Weight $\mu = 0.16$

From PDP8 investment allocations:

$$K_{b,ss} = \frac{2.4/10}{0.12} \approx 2.0 \text{ B USD (BESS)}$$

$$K_{g,ss} = \frac{14.9/10}{0.05} \approx 29.8 \text{ B USD (grid)}$$

With $\rho_f = 0.4$: $\mu \approx 0.16$. Reflects the *small but growing* share of private storage in Vietnam's current infrastructure.

Complementarity: $\rho_f = 0.40$

Transmission moves energy across *space*; storage moves it across *time*. When solar output is zero (night), infinite transmission capacity is useless. Hart (2019) and Hirth (2013) document strong complementarity empirically— $\rho_f = 0.4$ enforces this binding constraint.

Calibrated Parameters (I): Preferences, Production & Energy

Category	Parameter	Value	Source / Rationale
<i>Preferences & Production</i>	Discount factor β	0.990	Annualized real return $\approx 4\%$ (VGB)
	Capital share α	0.350	Labor income share = 65% (GSO 2024)
	Depreciation (prod.) δ_p	0.025	10% annually (standard)
	Labor elasticity σ_L	1.000	Frisch elasticity (NK literature)
	Energy-VA elasticity σ_E	0.600	Heutel (2012); energy–capital complementarity
<i>Energy & Reliability</i>	Reliability sensitivity ψ	2.000	Moment-matched: $u_{ss} = 97\%$ (EVN reports)
	Intermitt. persistence ρ_{ren}	0.850	Monsoon AR(1) (IEA 2023)
	Intermitt. volatility σ_{ren}	0.120	CV of monthly VN solar output (GSO 2024)
	Battery weight μ	0.160	PDP8 BESS/grid capital stock ratio
	Flexibility elasticity ρ_f	0.400	Complementarity (Hart 2019; Hirth 2013)
	Depreciation (battery) δ_b	0.030	12% annually (Li-ion lifecycle)
	Depreciation (grid) δ_g	0.0125	5% annually (transmission infrastructure)

Calibrated Parameters (II): Innovation, Shocks & External Sector

Category	Parameter	Value	Source / Rationale
<i>Learning-by-Doing</i>	Learning elasticity η_{bat}	0.100	18–22% Li-ion exp. curve (Kittner 2017; Way 2022)
	Signal transmission χ	1.0	Baseline: fully liberalized market
		0.3	Counterfactual: regulated tariff regime
<i>Battery Price Shock</i>	Persistence ρ_{bat}	0.900	AR(1) of lithium carbonate (BNEF 2023)
	Volatility σ_{bat}	0.080	Li-ion pack price survey (BNEF 2023)
<i>Fiscal Policy</i>	Grid invest. response ϕ_{grid}	1.500	Reactive lag structure (World Bank 2022)
<i>External Sector</i>	World interest rate $\bar{r}$	0.0101	$= 1/\beta - 1$ (quarterly)
	Debt premium ϕ_b	0.001	Schmitt-Grohé & Uribe (2003)
	SS net foreign assets $\bar{B}^*$	0	Balanced trade normalization

Calibration Validation: Model Fit to Vietnamese Data

Moment	Data	Model	Source
Investment rate (I/Y)	27–29%	27.2%	GSO 2024; targeted
Consumption share (C/Y)	64–66%	72.8% [†]	GSO 2024; model SS
Grid reliability (utilization)	95–98%	97.0%	EVN reports; targeted
Annualized real interest rate	$\approx 4\%$	4.04%	VGB average

[†] *Note:* Model abstracts from government consumption ($\approx 6\text{--}7\%$ of GDP). Adjusted private consumption share $\approx 66\%$ —consistent with data.

Reliability Parameter Validation

The steady-state utilization shortfall maps to Energy Not Served (ENS):

$$(1 - u_t) \approx \text{ENS}_t / \text{Demand}_t$$

Sensitivity Check

Results are robust for $\psi \in [1.5, 3.0]$ and $\mu \in [0.20, 0.35]$. The *qualitative* Agility Gap mechanism holds across the full parameter space tested.

Steady State Characterization

In the deterministic steady state (all shocks = 0, time subscripts dropped):

Steady-State Conditions

- ① $\alpha Y_{ss}/K_{p,ss} + 1 - \delta_p = 1/\beta$
- ② $u_{ss} = u^* = 0.97$; $A_{bat,ss} = 1$
- ③ $r_{ss}^* = \bar{r} = 1/\beta - 1$
- ④ $l_{grid,ss} = \delta_g K_{g,ss}$; $l_{bat,ss} = \delta_b K_{b,ss}$
- ⑤ $B_{ss}^* = 0$ (balanced trade)
- ⑥ $(1-\alpha)V_{ss} = \varphi C_{ss} L_{ss}^{1+\sigma_L}$

Required flexibility stock:
$$F_{ss} = -\frac{\phi_{int} \cdot \text{Vol}_{ren,ss}}{\psi} \ln(1 - u_{ss})$$

Variable	SS Value	Interpretation
Y_{ss}	0.868	Normalized
C_{ss}	0.631	72.8% of output
L_{ss}	0.330	Hours
$K_{p,ss}$	8.651	Productive capital
$K_{b,ss}$	0.190	Battery capital
$K_{g,ss}$	1.140	Grid capital
u_{ss}	0.970	Grid reliability
F_{ss}	0.526	Flexibility bundle
B_{ss}^*	0.000	Balanced trade
r_{ss}^*	0.0101	World rate

Log-Linearized System: Household & Production Block

Variables expressed as log-deviations: $\hat{x}_t \equiv \ln X_t - \ln X_{ss}$

Household Block

Capital Euler:

$$\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{\alpha r_{p,ss}}{1 + r_{p,ss}} \mathbb{E}_t \hat{Y}_{t+1} + \frac{\alpha r_{p,ss}}{1 + r_{p,ss}} \mathbb{E}_t \hat{K}_{p,t+1}$$

Labor supply:

$$\hat{V}_t = \hat{C}_t + (1 + \sigma_L) \hat{L}_t$$

Bond Euler:

$$\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{r_{ss}^*}{1 + r_{ss}^*} \mathbb{E}_t \hat{r}_{t+1}^*$$

Production & Reliability Block

CES output:

$$\hat{Y}_t = s_V^{\sigma^E} \hat{V}_t + (1 - s_V^{\sigma^E}) \cdot \underbrace{0}_{\bar{E} \text{ fixed}}$$

Value-added:

$$\hat{V}_t = \alpha (\hat{u}_t + \hat{K}_{p,t}) + (1 - \alpha) \hat{L}_t$$

Reliability (linearized around u^):*

$$\hat{u}_t = -\xi \hat{F}_t + \xi \hat{\text{Vol}}_{ren,t}$$

where $\xi \equiv \psi(1 - u_{ss}) / (\phi_{int} \cdot \text{Vol}_{ren,ss})$

Log-Linearized System: Energy, Innovation & External Block

Flexibility & Investment Block

CES flexibility (linearized):

$$\hat{F}_t = s_b(\hat{A}_{bat,t} + \hat{K}_{b,t}) + (1 - s_b)\hat{K}_{g,t}$$

Battery investment rule:

$$\hat{l}_{bat,t} = -\phi_{grid} \hat{u}_t - \hat{P}_{bat,t}$$

Grid investment rule:

$$\hat{l}_{grid,t} = -\phi_{grid} \hat{u}_t + \varepsilon_{I,t}$$

Innovation equation:

$$\hat{A}_{bat,t} - \hat{A}_{bat,t-1} = -\eta_{bat} \cdot \chi \cdot \hat{u}_t$$

External Sector Block

Current account:

$$\begin{aligned} B_t^* &= (1 + r_{ss}^*)B_{t-1}^* + Y_{ss} \hat{Y}_t - C_{ss} \hat{C}_t \\ &\quad - I_{p,ss} \hat{l}_{p,t} - P_{bat,ss} I_{bat,ss} (\hat{P}_{bat,t} + \hat{l}_{bat,t}) \\ &\quad - I_{grid,ss} \hat{l}_{grid,t} \end{aligned}$$

Interest premium:

$$\hat{r}_t^* = -\frac{\phi_b}{r_{ss}^*} B_t^*$$

“Twin drain”: an intermittency shock simultaneously reduces Y_t and raises battery import demand $I_{bat,t}$, creating a double hit on B^* .

Three Transmission Channels

Channel 1: The Reliability Wedge (Output Loss)

$$\varepsilon_t^{ren} \uparrow \Rightarrow \text{Vol}_{ren,t} \uparrow \Rightarrow u_t \downarrow \Rightarrow V_t \downarrow \Rightarrow Y_t \downarrow$$

Impact elasticity: $\partial \hat{Y} / \partial \varepsilon^{ren} \approx -\alpha \xi s_V^{(\sigma_E - 1) / \sigma_E} (1 - \omega_E) < 0$. Operates like a negative TFP shock: supply-side, persistent, amplified by CES complementarity $\sigma_E < 1$.

Channel 2: The Agility Gap (Asymmetric Investment)

$$u_t < u^* \Rightarrow I_{bat,t} \uparrow (\text{agile}); I_{grid,t} \uparrow \text{ but with 5–7 year construction lags } (\text{sluggish})$$

Battery investment responds contemporaneously; grid capital accumulates with a bureaucratic time-to-build lag captured in the capital accumulation equation.

Channel 3: Reliability-Gap Learning (Innovation)

$$u_t \downarrow \Rightarrow (u^* - u_t) / u^* \uparrow \Rightarrow \Delta A_{bat} \uparrow (\text{if } \chi > 0)$$

Reliability scarcity triggers endogenous cost reduction in storage. Conditional on χ : when regulated tariffs suppress scarcity signals, learning stalls. The “valley of death” for battery technology arises endogenously.

Solution Method: First-Order Perturbation (Dynare 6)

The model is solved using **first-order perturbation** around the deterministic steady state, implemented in *Dynare 6* (Adjemian et al. 2024):

State-Space Representation

$$\mathbf{s}_{t+1} = \mathbf{M} \mathbf{s}_t + \mathbf{N} \varepsilon_t$$

$$\mathbf{y}_t = \mathbf{P} \mathbf{s}_t$$

State vector:

$$\mathbf{s}_t = (\hat{K}_{p,t}, \hat{K}_{b,t}, \hat{K}_{g,t}, \hat{A}_{bat,t}, \hat{P}_{bat,t}, B_{t-1}^*, \hat{\varepsilon}_{ren,t})^\top$$

Blanchard-Kahn Conditions

3 eigenvalues outside the unit circle:

$$\lambda_1 = 1.033, \lambda_2 = -2.229, \lambda_3 = \infty$$

Matches **3 forward-looking variables**: C_t, Y_t, r_t^*

7 stable eigenvalues:

$$0, \approx 0, 0.90, 0.97, 0.98, 0.99, 0.99$$

$\Rightarrow$ Unique saddle-path stable solution $\checkmark$

Why first-order suffices: The exponential nonlinearity of u_t has been absorbed into the linearization coefficient ξ . The *Reliability Valley* (transition path) is analyzed via a separate deterministic simulation to capture nonlinear effects beyond the linearized neighborhood.

Baseline Results: Steady State & FEVD Overview

Forecast Error Variance Decomposition (40Q)

Variable	ε_{ren}	ε_{bat}	ε_I
Output (Y)	79%	16%	5%
Reliability (u)	85%	12%	3%
Consumption (C)	11%	80%	9%
Battery invest.	1%	100%	<1%
Grid invest.	14%	1%	85%

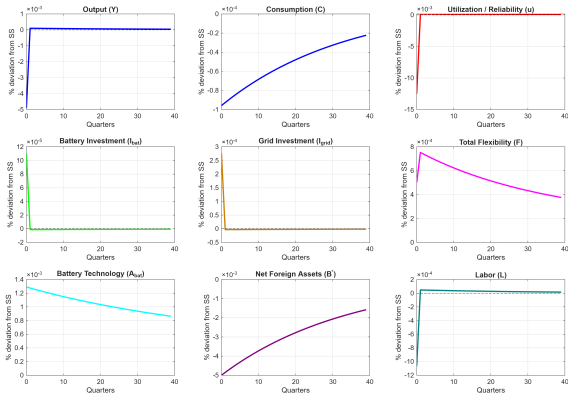
Three Structural Findings from FEVD

- 1 **Intermittency dominates** output (79%) and reliability (85%)—validates the model's core reliability channel
- 2 **Battery investment** is 100% driven by global price shocks, not domestic policy—Vietnam is a price-taker
- 3 **Consumption** is dominated by battery price shocks (80%), not intermittency—household smooths domestic shocks via borrowing but is exposed to import price volatility

Note: The B^* shares exceed 100% due to negative covariance: intermittency drains the current account (output falls) *and* battery price shocks raise import costs simultaneously.

Impulse Response: Renewable Intermittency Shock

Impulse Responses to Renewable Intermittency Shock (ϵ_{ren})



Three Stages

Stage 1 (impact):

$$u_t \downarrow -0.013\%$$

$$Y_t \downarrow -0.005\%$$

Reliability penalty activates

Stage 2 (Q1–Q10):

$$I_{bat} \uparrow +0.0001\% \text{ (agile)}$$

$$I_{grid} \uparrow +0.0003\% \text{ (sluggish)}$$

Agility gap visible

Stage 3 (Q10+):

$$A_{bat} \uparrow +0.11\% \text{ by Q10}$$

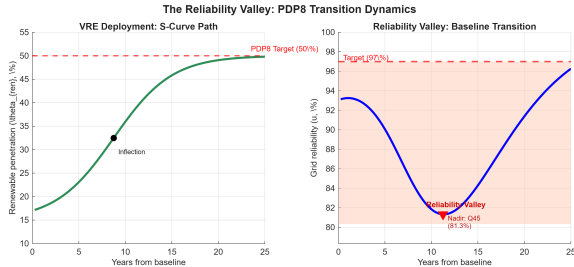
Endogenous LBD activated

Gradual reliability recovery

One standard deviation shock to renewable intermittency ($\sigma_{ren} = 0.12$). All variables:

% deviation from steady state over 40 quarters.

The Reliability Valley: PDP8 Transition Path (Preliminary)



Deterministic simulation: VRE penetration 15%→50% over 100 quarters (S-curve).

Reliability falls as VRE outruns flexibility accumulation.

Valley Statistics

Nadir at:	Quarter 45
Min. reliability:	81.3%
Below target:	15.7 pp
Implied ENS:	≈ 1,640 hrs/yr

Three Causes

- ① Front-loaded VRE (FiT boom)
- ② Lagged grid (5–7 yr permitting)
- ③ Battery supply constraints

Full welfare analysis and policy counterfactuals in progress—to be presented at final defense.

Work in Progress

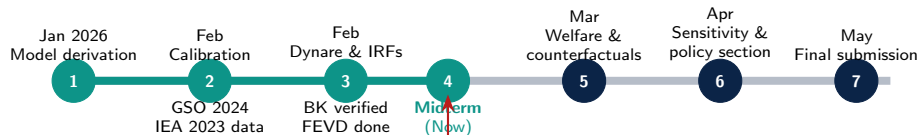
Completed ✓

- Full model derivation (16-equation equilibrium)
- Analytical steady-state characterization & solver
- Dynare implementation & Blanchard-Kahn verification
- Full calibration (structural + reliability + LBD blocks)
- Impulse response functions (3 shocks)
- Forecast Error Variance Decomposition (40Q horizon)
- Deterministic transition simulation (Reliability Valley)
- Log-linearization & transmission channel analysis

In Progress / Remaining

- **Welfare analysis:** consumption-equivalent variation $\Lambda(\chi)$
- **Counterfactual experiments:**
 - Battery subsidy ($P_{bat} = 0.8$)
 - Accelerated grid ($\phi_{grid} = 3.0$)
 - Signal suppression ($\chi = 0.3$)
- **“Perfect Storm” scenario:** joint $\varepsilon_{ren} + \varepsilon_{bat}$ simulation
- **Sensitivity analysis:** $\psi, \mu, \beta, \phi_{grid}$
- **Formal propositions:** prove $\Lambda(\chi)$ monotonicity and CES supermodularity
- **Policy section:** map $\chi \rightarrow$ concrete EVN reform agenda
- **LaTeX manuscript:** finalize bibliography & appendix proofs

Completion Timeline



At midterm: model fully solved, calibrated, and initial quantitative results obtained. Remaining work is welfare analysis, counterfactuals, and manuscript finalization.

Summary: Three Novel Contributions to the Literature

1 Endogenous Grid Reliability as a Macro Friction

Capital utilization $u_t = 1 - \exp(-\psi F_t / \phi_{int} \text{Vol}_{ren,t})$ is determined endogenously by the stock of flexibility assets relative to VRE volatility. This transforms an engineering concept into a macroeconomic wedge in the aggregate production function—a channel absent from all extant E-DSGE models.

2 Reliability-Gap-Induced Learning-by-Doing

Battery technology evolves endogenously: $\Delta A_{bat} = \eta_{bat} \cdot \chi \cdot (u^* - u) / u^*$. Unlike DTC models that require explicit R&D sectors, innovation here responds to *deployment-driven scarcity*. The regulatory parameter χ maps directly to observable policy instruments (tariff pass-through, capacity market design).

3 Identification of the Structural Agility Gap

Private batteries (agile, price-driven) and public grid capital (sluggish, rule-based) are formally distinguished as *complementary* inputs in a CES flexibility bundle ($\rho_f = 0.4$). This generates a predictable mid-transition reliability valley—a quantitative prediction unavailable in models that treat flexibility as homogeneous.

Thank You

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Midterm Progress Report

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Renewable Intermittency · Energy Storage · Vietnam PDP8

Code repository: `vietnam_dsge.mod` + `run_dynare.m` (MATLAB/Dynare 6)

Appendix: Full System of Log-Linearized Equations

The complete 16-equation linearized system (hat notation: $\hat{x}_t \equiv \ln X_t - \ln X_{ss}$):

① Capital Euler: $\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{\alpha r_{p,ss}}{1+r_{p,ss}} (\mathbb{E}_t \hat{Y}_{t+1} - \mathbb{E}_t \hat{K}_{p,t+1})$

② Bond Euler: $\hat{C}_t = \mathbb{E}_t \hat{C}_{t+1} - \frac{r_{ss}^*}{1+r_{ss}^*} \mathbb{E}_t \hat{r}_{t+1}^*$

③ Labor: $\hat{V}_t = \hat{C}_t + (1 + \sigma_L) \hat{L}_t$

④ CES output: $\hat{Y}_t = s_V^{\sigma} \hat{V}_t$

⑤ Value-added: $\hat{V}_t = \alpha(\hat{u}_t + \hat{K}_{p,t}) + (1 - \alpha) \hat{L}_t$

⑥ Reliability: $\hat{u}_t = -\xi \hat{F}_t + \xi \hat{V}_{ol_{ren},t}$

⑦ Flexibility: $\hat{F}_t = s_b(\hat{A}_{bat,t} + \hat{K}_{b,t}) + (1 - s_b) \hat{K}_{g,t}$

⑧ Battery invest: $\hat{I}_{bat,t} = -\phi_{grid} \hat{u}_t - \hat{P}_{bat,t}$

⑨ Battery capital: $\hat{K}_{b,t+1} = (1 - \delta_b) \hat{K}_{b,t} + \frac{I_{bat,ss}}{K_{b,ss}} \hat{I}_{bat,t}$

⑩ Learning: $\hat{A}_{bat,t} - \hat{A}_{bat,t-1} = -\eta_{bat} \cdot \chi \cdot \hat{u}_t$

⑪ Grid invest: $\hat{I}_{grid,t} = -\phi_{grid} \hat{u}_t + \varepsilon_{I,t}$

⑫ Grid capital: $\hat{K}_{g,t+1} = (1 - \delta_g) \hat{K}_{g,t} + \frac{I_{grid,ss}}{K_{g,ss}} \hat{I}_{grid,t}$

⑬ Prod. capital: $\hat{K}_{p,t+1} = (1 - \delta_p) \hat{K}_{p,t} + \frac{I_{p,ss}}{K_{p,ss}} \hat{I}_{p,t}$

⑭ Current account: [linearized B_t^* equation]

⑮ Interest premium: $\hat{r}_t^* = -\frac{\phi_b}{r_{ss}^*} B_t^*$

⑯ Battery price AR(1): $\hat{P}_{bat,t} = \rho_{bat} \hat{P}_{bat,t-1} + \varepsilon_t^{bat}$

Appendix: Welfare Cost Formula

Consumption-Equivalent Variation Λ (Lucas 1987)

The welfare cost of intermittency is defined as the permanent percentage reduction in steady-state consumption that leaves the household indifferent between living in the stochastic economy vs. the deterministic steady state:

$$\Lambda = 1 - \exp \left[\frac{1 - \beta}{1 - \beta^{T+1}} \sum_{t=0}^T \beta^t \log \left(\frac{C_t^{shock}}{C_{ss}} \right) \right]$$

where $T = 40$ quarters (10 years) and C_t^{shock} follows the IRF path.

Proposition: $\Lambda(\chi)$ Strictly Decreasing

Higher signal transmission $\chi \Rightarrow$ faster A_{bat} improvement $\Rightarrow$ higher $F_t \Rightarrow$ higher $u_t \Rightarrow$ higher $Y_t \Rightarrow$ higher $C_t \Rightarrow$ lower Λ .

Proof: by monotone comparative statics on the

Preliminary Result

χ	Λ	Δ from baseline
1.0 (baseline)	0.000054%	—
0.5 (partial)	0.000063%	+17%
0.0 (suppressed)	0.000073%	+35%